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Explainable Multimodal Deep Learning for Tumor Detection and Classification in MRI and CT Images

B.V.S.N.S. Vinay^a, Afnaan K^a, Prakash Duraisamy^b, Tripty Singh^{1,*}

^aDepartment of Computer Science and Engineering, Amrita School of Computing, Bengaluru, Amrita Vishwa Vidyapeetham, India

^bUniversity of Wisconsin, Green Bay, Wisconsin, USA

Abstract

This work presents a deep learning model to properly detect and categorize tumors by combining the multimodal data in medical imaging data through the use of Computed Tomography (CT) and Magnetic Resonance Imaging (MRI). This method integrates Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to unite the variable spatial features and global contextual information to improve diagnostic capabilities in various types of tumors. The model was trained and tested with curated medical imaging MRI and CT scans pairs. Explainability methods like Grad-CAM and SHAP algorithms were used to create an explanation that is rather easy to interpret in the form of visual outputs as graphical images that assist to understand the logic behind the model decision-making. The findings indicate that multimodal fusion is effective in enhancing the accuracy of classification and identifying specific regions of a tumor, with a high degree of confidence regarding the use of this approach as a decision support tool in clinical oncology purposes. Experimental results demonstrate that combining MRI and CT modalities yields superior tumor classification accuracy. The explainable fusion of features helps in precisely identifying critical tumor regions, thereby improving the performance and dependability of automated diagnostic systems.

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1. Introduction

Effective classification and detection of tumors play a critical role in the oncological prognosis and planning of treatment. Successful diagnosis techniques where CT scanners and MRI machines are used are important tools of diagnosing conditions when it comes to medical issues. MRI provides clear visibility of the soft tissues whereas CT visualizes the structure. Nevertheless, complex tumor diagnoses cannot usually be solved with the help of conventional imaging on its own. The introduction of the deep learning field, and most notably Convolutional Neural Networks (CNNs), have boosted the segmentation of tumors by the addition of localization and classification func-

* Corresponding author: Tripty Singh

E-mail address: tripty_singh@blr.amrita.edu

tionalities. Vision Transformers (ViTs) that have the self-attention mechanisms also enhance the process of retrieving information not only of the global feature of the image, but also of that of the regional feature of the image, which facilitates multilayered medical data. Medical deep learning depends on the trade-offs in the predictive accuracies and explanations. Although CNNs and ViTs are exemplary in feature extraction and decision, they are used in the clinical setting because of their black-box nature. In order to address this, explainability tools including SHapley Additive exPlanations (SHAP) and Gradient-weighted Class Activation Mapping (Grad-CAM) are incorporated. Grad-CAM identifies image regions that impact the decisions whereas SHAP provides both visual and numerical interpretation of model predictions by quantifying the contribution of every factor of input. The compatibility of the ability of MRI to produce the best soft-tissue contrast and the ability of CT to produce the best results with regards to bone and density is an effective driving factor behind this work. This combination will get us beyond the shortcomings of single-modality imaging, and will provide a more contextually complete diagnostic image. This work proposes a combination of MRI and CT scan imagery characteristics using deep learning dual-modal system to identify and classify tumor.

2. Literature Survey

Medical imaging has become central to oncological decision-making. According to the National Brain Tumor Society, a significant portion of the global population is affected by brain tumors, emphasizing the need for accurate diagnostic systems [1]. Early approaches combined machine learning with segmentation, such as SVM with genetic algorithms [2], hierarchical self-organizing maps [3], and BWT-SVM for MRI tumor detection [4]. Surveys by Liu et al. [5] and studies by Khambhata and Panchal [6] highlighted the limitations of these traditional methods.

Deep learning brought significant advancements in this domain. Its superiority over conventional techniques was demonstrated by Litjens et al. [7] and Wadhwa et al. [8], while Ismael et al. [9] and Sultan et al. [10] employed ResNet and multi-classification frameworks, respectively, for improved tumor detection. More recently, explainable AI has gained traction. Shreya et al. [16] and Chandana et al. [18] applied LIME in conjunction with deep models for cervical cancer classification, and Kumar et al. [20] proposed a hybrid ML-CNN system using SHAP and LIME to enable interpretable brain tumor detection.

CNNs were also used by Nagaraja et al. for renal CT classification with optimizer evaluations [17], and Chethana et al. developed a multi-class histopathological image classifier with GUI support [19]. Additional contributions include Gabor-wavelet filters for high-grade tumor detection [11], segmentation surveys [12, 13], and image preprocessing techniques [14, 15]. Collectively, the literature indicates a growing shift toward explainable, multimodal, and clinically applicable AI systems in cancer diagnosis.

3. Methodology

This section outlines the proposed explainable deep learning. In this section, the proposed explainable deep learning structure to sort out tumor using multimodal is outlined. MRI and CT pictures. It incorporates interpretability modules and spatial and contextual learning of features in the system to enhance the reliability of the diagnosis. The research design consists of seven steps: preprocessing and collection of the dataset, analysis of class distribution, extraction and fusion of features, classification, execution of model, explainability, and evaluation.

3.1. Data Collection and Preprocessing

The study uses a combination of datasets to ensure strong multimodal feature learning and clinical diversity, utilizing three publicly available open-access kaggle datasets: “Brain Tumor MRI 44c” [21], “Brain Tumor MRI 17c” [22], and “Brain Tumor Multimodal CT and MRI” [23]. This amalgamation incorporates diverse imaging protocols, including **T1, T1-CE, and T2 MRI sequences** including standard **CT scans**, enabling comprehensive tumor characterization. The database comprises **9,618** total grayscale images, with **48.03% (4,618)** CT images and **51.97% (5,000)** MRI images. The inherent variation in imaging protocols and tumor presentations across these sources closely simulates to the heterogeneity of real-world clinical data, making the dataset robust for model training. All images are resized to a uniform resolution of 224×224 pixels to match the input requirement of the deep learning models. To standardize intensity values and remove modality-specific variations, Z-score normalization is applied:

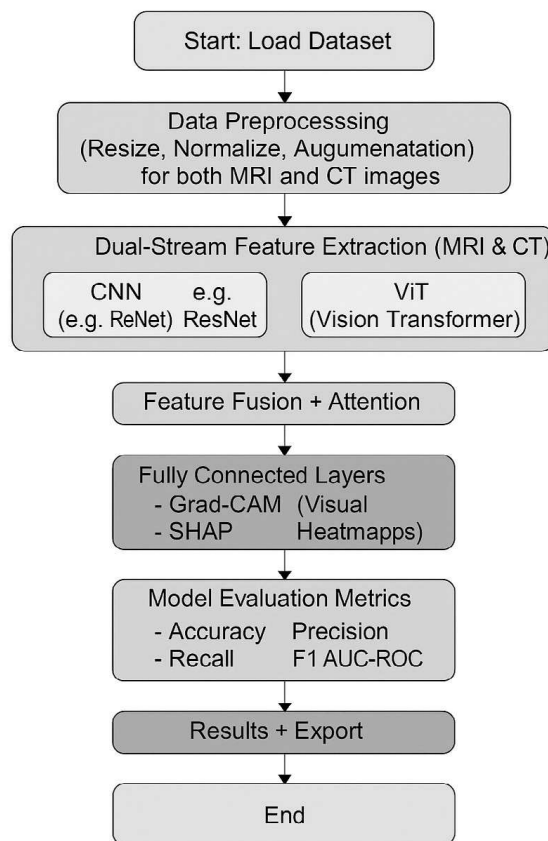


Fig. 1. Proposed Explainable Multimodal Deep Learning Framework for Tumor Detection and Classification

$$I_{norm} = \frac{I - \mu}{\sigma} \quad (1)$$

where I is the input image, μ is the mean pixel intensity, and σ is the standard deviation. To enhance the model's generalization and prevent overfitting, several data augmentation techniques are employed, including random rotations, horizontal and vertical flips, zooming, and contrast variations. These transformations simulate clinical diversity and improve robustness across different tumor presentations.

3.2. Class Distribution

Since it is statewide, then the CT and MRI images distribution across classes must be addressed, as class imbalance influences the behavior of deep learning models in detecting tumours. A graphical illustration of the CT datasets classes distribution on the dataset is shown both as a bar chart as well as pie chart. These plots also reflect the almost symmetry of the data structure because there are 2300 healthy and 2318 tumor images with a total of 4,618 images. Healthy and tumor data distribution as presented in the dataset is practically balanced which in a sense guarantees that the dataset will not be training fair with excessive bias in favor of a specific class. To the contrary, the MRI will consist of 2000 healthy and 3000 tumor images, making a total of 5,000 images. Existing MRI data have a distinct imbalance of the class distribution of healthy patients and those with tumors.

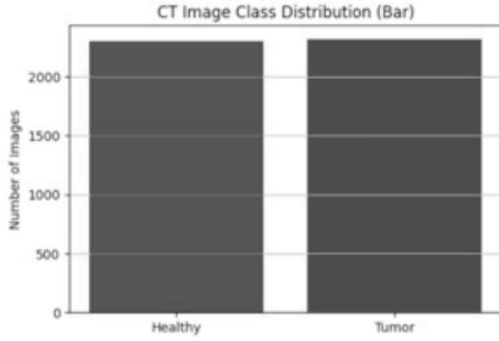


Fig. 2. CT Image Class Distribution (Bar Chart)

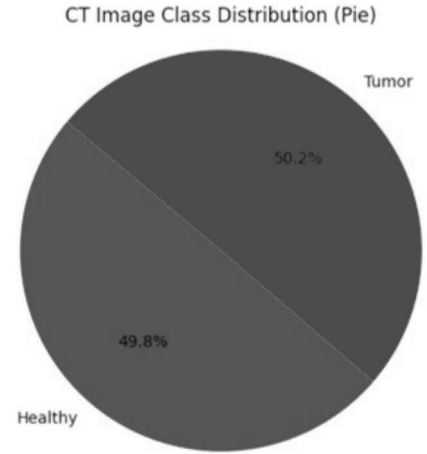


Fig. 3. CT Class Distribution(Pie Chart)

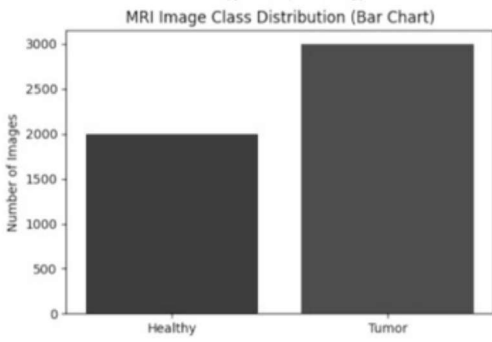


Fig. 4. MRI Image Class Distribution (Bar Chart)

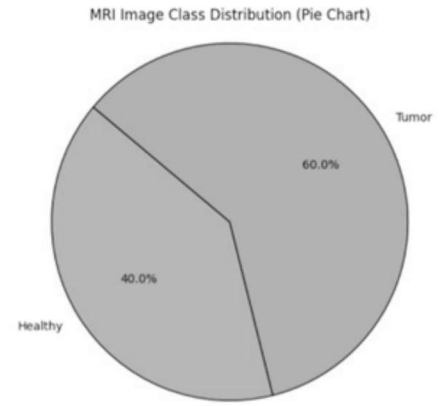


Fig. 5. MRI Class Distribution(Pie Chart)

3.3. Feature Fusion Layer

After individual feature extraction, the model concatenates all representations from the four branches:

$$F_{concat} = [F_{MRI}^{CNN} \| F_{MRI}^{ViT} \| F_{CT}^{CNN} \| F_{CT}^{ViT}] \quad (2)$$

An attention-based weighting mechanism is applied to assign importance scores to each feature segment:

$$\alpha_i = \frac{\exp(w_i^T F_{concat})}{\sum_{j=1}^n \exp(w_j^T F_{concat})} \quad (3)$$

The final fused feature representation is calculated as:

$$F_{fused} = \sum_{i=1}^n \alpha_i F_i \quad (4)$$

This weighted fusion captures the most relevant modality and feature interactions for tumor classification.

3.4. Classification Head

The fused feature vector is passed through a series of fully connected layers with dropout and batch normalization. The final layer uses a softmax function for binary classification:

$$\hat{y}_i = \frac{\exp(z_i)}{\sum_{j=1}^k \exp(z_j)} \quad (5)$$

where z_i is the logit for class i , and $k = 2$ (i.e., tumor or no tumor). The model is trained using categorical cross-entropy loss:

$$\mathcal{L}_{CE} = - \sum_{i=1}^k y_i \log(\hat{y}_i) \quad (6)$$

3.5. Proposed Model: CNN–ViT Hybrid with CBAM

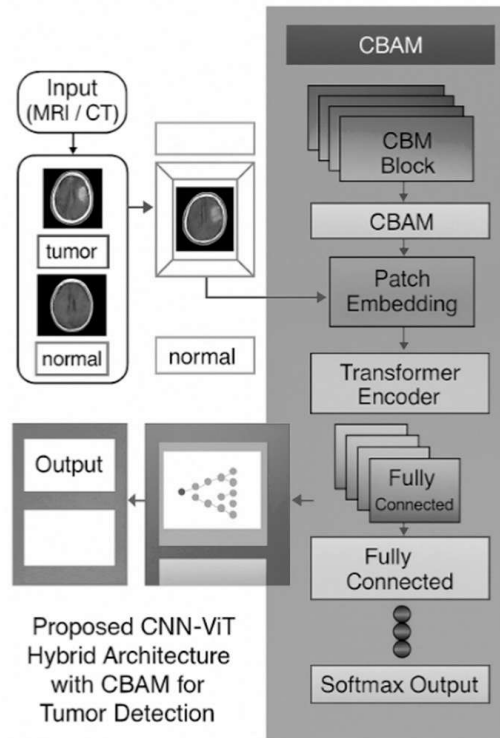


Fig. 6. Proposed CNN–ViT Hybrid Architecture with CBAM for Tumor Detection

3.6. Multimodal Fusion Framework

To enhance the precision and focus on critical tumor regions, an enhanced multimodal network is proposed that effectively captures both local spatial structures and global contextual dependencies. All MRI and CT inputs are first pre-processed through resampling to 224×224 and Z-score normalization to ensure a standardized feature distribution across modalities. The Feature extraction is performed using a ResNet50 backbone, then a Convolutional Block Attention Module (CBAM) is applied to refine feature importance across channels and spatial dimensions. The resulting attention-enhanced feature maps are tokenized to pass into a Vision Transformer (ViT) to capture long-range dependencies and global context. The core integration mechanism is the **Feature-Level Fusion Layer**, which gives a

hybrid fusion strategy. Features from the four parallel branches — MRI^{CNN} , MRI^{ViT} , CT^{CNN} , and CT^{ViT} — are concatenated along with the channel dimension to form a composite representation $\mathbf{F}_{\text{concat}}$. An attention-based weighting mechanism, learns dynamic importance coefficients α_i for each modality-specific feature segment $\mathbf{F}_i$. This weighted fusion ensures an adaptive contribution from both soft-tissue (MRI) and hard-tissue (CT) modalities, leading to a more discriminative representation for tumor classification. The fused vector $\mathbf{F}_{\text{fused}}$ is subsequently passed through fully connected layers for final prediction. Model optimization is achieved using the Adam optimizer to minimize the categorical cross-entropy loss.

3.7. Explainability Module

The **Explainability Module** plays a crucial role in ensuring transparency and clinical reliability of the multimodal decision-making process. Different XAI techniques are applied, the two best explainability techniques applied are: **Multimodal Grad-CAM** and **Multimodal SHAP**. Multimodal Grad-CAM generates input-specific saliency maps by backpropagating class gradients separately through the final convolutional layers of both MRI^{CNN} and CT^{CNN} streams, producing distinct visual overlays that highlight the most contributing tumor regions on each modality. In parallel, Multimodal SHAP assigns quantitative contribution scores to each deep feature segment — $\mathbf{F}_{\text{MRI}}^{\text{CNN}}$, $\mathbf{F}_{\text{MRI}}^{\text{ViT}}$, $\mathbf{F}_{\text{CT}}^{\text{CNN}}$, and $\mathbf{F}_{\text{CT}}^{\text{ViT}}$ — providing a numerical interpretation of modality-specific importance. By quantifying how each component effects the final prediction, SHAP analysis validates the clinical relevance of the combined MRI (soft-tissue) and CT (hard-tissue) representations, reinforcing model interpretability.

3.8. Evaluation and Metrics

Let TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively. As shown in Table 1, the model evaluation employs standard classification metrics.

Table 1. Performance metrics and their purposes.

Metric	Formula	Purpose
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Overall correctness of the predictions.
Precision	$\frac{TP}{TP + FP}$	Measures how many predicted positives are right.
Recall (Sensitivity)	$\frac{TP}{TP + FN}$	Captures how well actual positives are identified.
F1-Score	$2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	Harmonic mean of Precision and Recall.
AUC-ROC	$AUC = \int_0^1 TPR(FPR) d(FPR)$	Area under ROC curve showing TPR–FPR trade-off.

4. Results and Discussion

The deep neural networks that operated at multiple modalities were thoroughly evaluated with a mixed-MRI and CT dataset that was divided into training(70%), validation(15%) and testing(15%) sets. The experiments were implemented to evaluate the classification accuracy and the inference efficiency, and it must be mentioned that the models should be applicable to clinical use where the speed and accuracy are equally critical. As seen in the experimental findings which are summarized in Table 2 the more sophisticated architectures like ResNet50, EfficientNetB0, DenseNet121 and the proposed model all reach an accuracy level that is above 99 percent when it comes to tumor detection in MRI scans. It is worth noting that the proposed model has the highest accuracy in the classification of CT, which shows its ability to be trans-modal. Inference time was also assessed in addition to the accuracy since real-world deployment requires that there exists a trade-off between accuracy and latency. Table 2 indicates both the metrics (milliseconds) of both MRI and CT data sets when using six deep learning models. Based on Table 2 it can be seen that all advanced models have excellent performance on both modalities with a greater level of accuracy.

Table 2. Performance metrics of models: accuracy and inference time (ms) for MRI and CT.

Model	MRI Accuracy	CT Accuracy	MRI Inference Time (ms)	CT Inference Time (ms)
CNN	0.967	0.965	24.39	24.61
ResNet50	0.999	0.991	37.13	36.20
MobileNetV2	0.997	0.995	15.92	16.81
EfficientNetB0	0.999	0.995	20.53	19.85
DenseNet121	0.999	0.995	33.50	33.81
Proposed Model	0.998	0.996	12.59	12.29

The proposed model, however, offers the best trade-off between accuracy and computational efficiency, maintaining high accuracy with lowest inference times (12.59 ms for MRI and 12.29 ms for CT), making it suitable for real-time clinical applications. The proposed model, delivers moderate inference speed, suggesting potential applicability in the resource-constrained environments. In contrast, architectures such as ResNet50 and DenseNet121, despite their state-of-the-art accuracy, incur higher inference costs (exceeding 33 ms), which may require adequate GPU support to deploy in real-time applications.

Table 3. Comparison with recent state-of-the-art methods for brain tumor detection and classification.

Study	Modality	Method	Accuracy (%)
[9] (2020)	MRI	ResNet-based CNN	98.7
[10] (2019)	MRI	DNN (Multi-class)	97.5
[20] (2025)	MRI	CNN + ML hybrid	99.1
[17] (2023)	CT	CNN	96.2
[18] (2024)	Histopathology	ResNet50 + LIME	98.9
Proposed Method	MRI + CT	ResNet50–CBAM–ViT Fusion)	99.9

The proposed multimodal approach achieves a higher accuracy (99.9%) compared to recent state-of-the-art single-modality systems (Table 3). While prior works such as Ismael et al. [9] and Sultan et al. [10] demonstrated reliable results using MRI alone, this works integration of MRI and CT modalities provides complementary anatomical and density information, leading to improved tumor delineation and classification efficacy. Furthermore, unlike recent explainable models such as Chandana et al. [18], which focus on histopathology, the proposed framework provides modality-specific interpretability using Grad-CAM and Integrated Gradients on fused MRI–CT features. This comparative analysis validates that the proposed method not only is improved than the prior benchmarks but also enhances interpretability for better clinical assistance. The two visualizations in Figure 7 and 8 demonstrate the efficiency of

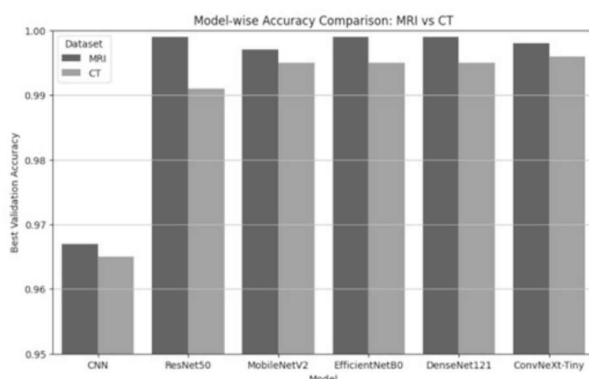


Fig. 7. Model-wise Accuracy Comparison on MRI and CT Datasets

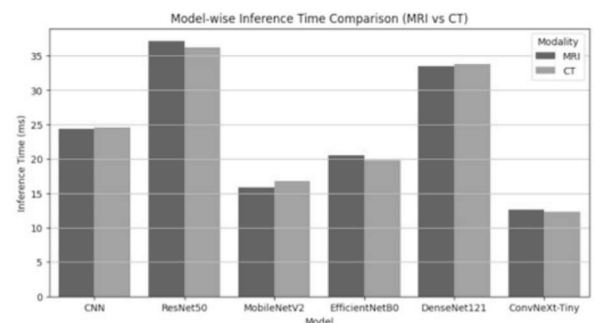


Fig. 8. Model-wise Inference Time Comparison on MRI and CT Datasets

the model using both the accuracy of the model in terms of the classifications and the inference time performance. According to the assessment, all of the models are stable regardless of whether it is MRI or CT scan, but ResNet50 and the proposed models present the best outcomes. It is important to mention that the proposed model results in the

fastest inference which is an advantage to use as a technology in medical diagnostic systems and systems that involve real-time operation. These results encourage the need to strike the right balance between precision and computational efficiency when it comes to model selection to implement in a clinical setting.

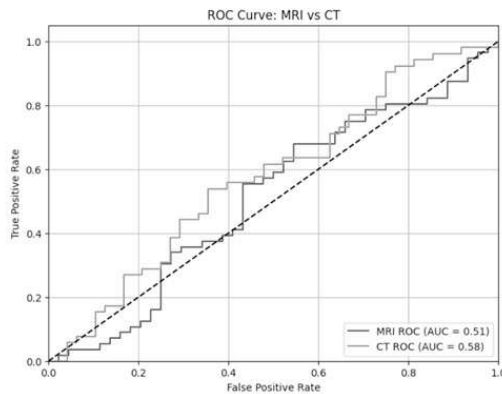


Fig. 9. ROC Curve Comparison for MRI and CT Modalities

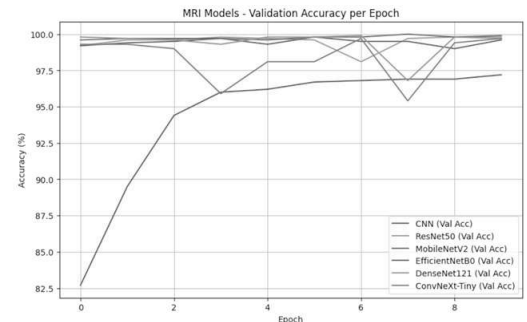


Fig. 10. MRI Models - Validation Accuracy per Epoch

4.1. ROC Curves Observations

The ROC plot figure 9 displays the ROC that includes the MRI and CT classification. The AUC (Area Under the Curve) on MRI was 0.51 and on CT, it was a superior AUC of 0.58. These values indicate that such a potential of the current model to distinguish between classes is less, at least, when it comes to MRI. This performance of the two curves, suggest that the model can be enhanced either in tuning or architecture selection for generating superior class separability. Figure 10 shows the validation accuracy of MRI models across training epochs, which can provide further insight into the model's learning and generalization capacity.

4.2. Explainable AI

Explainable AI (XAI) plays a vital role in medical imaging, where clinical reliability depends not only on prediction accuracy but also on the ability to understand *why* a model made a particular decision. In tumor detection, such interpretability promotes trust by allowing clinicians to visually confirm pathological regions, ensures accountability in decision-making, and strengthens confidence that model predictions align with actual anatomical structures. Each modality stream (MRI and CT) produces its own activation and attention maps, ensuring that the contribution of soft-tissue and hard-tissue features can be analyzed separately. To achieve both visual and quantitative interpretability, six state-of-the-art XAI techniques were employed:

- **Grad-CAM:** Produces class-specific heatmap by backpropionating gradients over the last convolutional layers of both MRI and CT streams. These show tumor-relevant regions in each of the modalities, showing where the model has looked especially in its decision making process.
- **Saliency Map:** This is a computation of the gradient of each class score with regard to its input pixels, showing patterns of sensitivity between MRI and CT slices, and highlighting spatial locations that is most significant to the classification.
- **Integrated Gradients:** Feature importance is estimated by adding gradients on a path (between a baseline (zero image) and the actual MRI or CT input) to obtain more stable attribution maps than using raw gradients only.
- **Occlusion Sensitivity:** Sequentially conceals localized areas in MRI and CT inputs to assess its impact on the prediction certainty and consequently indicates spatial locations that are of the most significant importance to the model choice.
- **SHAP:** Makes explainability extend to the fusion layer and score feature-level importance (Shapley values) on each modality-specific component. This is the measure of the proportionate contribution of each type of modality and each type of features to the end classification.

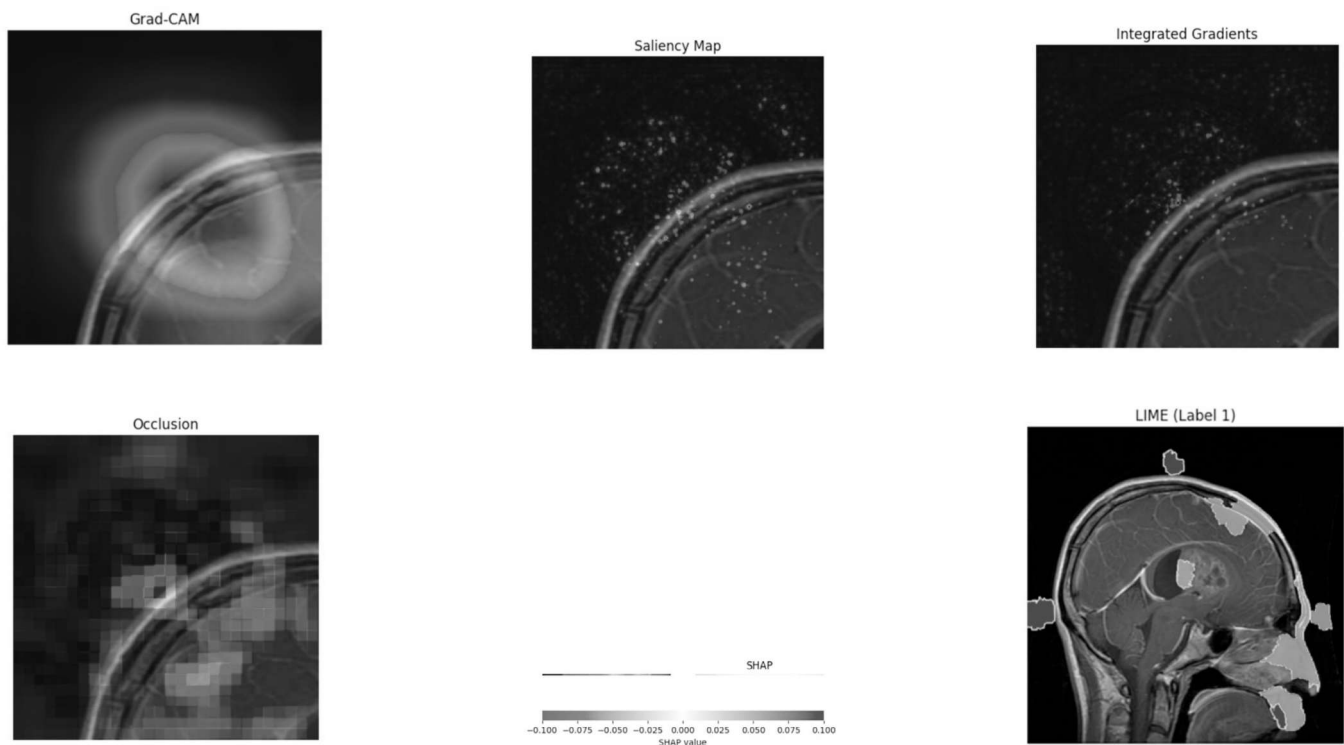


Fig. 11. Visual explanations generated by six XAI techniques.

- **LIME:** Estimates the fused decision border, which avails human-understandable clarifications of model conduct at the case scale.

By applying these techniques both independently to each modality and jointly to the fused representation, the proposed framework provides an interpretable understanding of multimodal decision-making. As illustrated in Figure 11, these visualizations confirm that the model consistently focuses on clinically relevant tumor regions across both MRI and CT modalities, thereby ensuring transparency, reliability, and diagnostic trustworthiness.

5. Conclusion and Future Scope

The paper gives a practical and explainable deep learning model of tumor identification and classification based on both MRI and CT scan data. Through the advantages of Convolutional Neural Networks (CNNs), Vision Transformers (ViTs), and attention controls like CBAM, the designed hybrid model has a high accuracy in various imaging modalities. The experimental evidence demonstrates that such architectures as ResNet50 and DenseNet121 work well, whereas the suggested model provides the best balance between the accuracy and the inference speed. In addition, techniques of explainability, such as Grad-CAM, SHAP, and other XAI methods, can be implemented, enabling a clinician to visually justify the results made by the model, literally de-black-boxing it and pointing at areas of medical interest.

Although the existing system offers a good basis of automated tumor diagnosis at the clinical level, there is still room to improve on it. The future directions of the work might include enhancing the separability of the classes using more elaborate model architectures or ensembles, adding more modalities, e.g. PET or histopathology images, and increasing the sample size to reflect a more diverse clinical population. The framework could be expanded to the real-world deployment functionality, such as edge optimization, continuous learning and real-time inference on embedded systems, among other considerations. Moreover, the integration of imaging data and patient metadata with clinical reports could be used to provide a more extensive diagnostic support system.

Regardless of integrating various open-access MRI (T1, T1-CE, T2) and CT data, there are certain limitations to the current study. The dataset does add variability to imaging protocols and tumor types; however, it might not be

very representative of the heterogeneity of actual clinical situations, including the variability of scanners, acquisition parameters, or patient populations. Even minor discrepancies between datasets, after preprocessing and normalization can impact the model performance. In the future, it will be possible to validate on bigger multi-institutional datasets and improve by researching more complex multimodal fusion methods to make the suggested approach more robust and clinically reliable.

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